

Arjuna JEE 2025

Maths

Sequence & Series

DPP: 2

- Q1** If the n^{th} term of an AP be $(2n - 1)$, then the sum of its first n terms will be
 (A) $n^2 - 1$
 (B) $(2n - 1)^2$
 (C) n^2
 (D) $n^2 + 1$
- Q2** If the sum of n terms of an AP is $2n^2 + 5n$, then the n^{th} term will be
 (A) $4n + 3$ (B) $4n + 5$
 (C) $4n + 6$ (D) $4n + 7$
- Q3** If the sum of $2n$ terms of an AP 2, 5, 8, 11, ... is equal to the sum to n terms of an AP 57, 59, 61, 63, ..., then n is equal to
 (A) 10 (B) 11
 (C) 12 (D) 13
- Q4** If $a_1, a_2, a_3 \dots$ are in AP and $a_1 + a_7 + a_{16} = 40$ then $S_{15} =$
 (A) 100 (B) 150
 (C) 200 (D) 250
- Q5** Three numbers are in A. P such that their sum is 18 and sum of their square is 158. The greatest among them is
 (A) 10 (B) 11
 (C) 12 (D) 14
- Q6** The interior angel of a polygon are in A. P. The smallest angel is 120° and the common difference is 5° . The numbers of sides of the polygon are
 (A) 16
 (B) 9
 (C) 9 and 16
 (D) None of these
- Q7** The sum of all two digit natural numbers which leave a remainder 5 when they are divided by 7 is equal to
 (A) 715 (B) 702
 (C) 615 (D) 602
- Q8** If 100 times the 100^{th} term of an AP with non-zero common difference equals to 50 times its 50^{th} term, then the 150^{th} term of an AP is
 (A) -150
 (B) 150 times its 50^{th} term
 (C) 150
 (D) Zero
- Q9** Which term of the sequence $(-8 + 18i), (-6 + 15i), (-4 + 12i), \dots$ is purely imaginary?
 (A) 7^{th}
 (B) 6^{th}
 (C) 5^{th}
 (D) 4^{th}
- Q10** If a, b, c are in AP, then the value of $(a + 2b - c)(2b + c - a)(a + 2b + c)$ is
 (A) $4abc$
 (B) $8abc$
 (C) $3abc$
 (D) $16abc$
- Q11** The ratio of the sums of first n even numbers and n odd numbers will be
 (A) $1 : n$



- (B) $(n + 1) : 1$
 (C) $(n + 1) : n$
 (D) $(n - 1) : 1$
- Q12** If $1, \log_9(3^{1-x} + 2), \log_3(4 \cdot 3^x - 1)$ are in AP, then x is equal to
 (A) $\log_3 4$
 (B) $1 - \log_3 4$
 (C) $1 - \log_4 3$
 (D) $\log_4 3$
- Q13** If S_n denotes the sum of first n terms of an A.P., and $S_{2n} = 3S_n$ then $\frac{S_{3n}}{S_n}$ is equal to
 (A) 4 (B) 5
 (C) 6 (D) 7
- Q14** If $a_1, a_2, a_3, \dots$ are in A.P. such that $a_1 + a_5 + a_{10} + a_{15} + a_{20} + a_{24} = 225$, then $a_1 + a_2 + a_3 + \dots + a_{23} + a_{24}$ is equal to
 (A) 909 (B) 75
 (C) 750 (D) 900
- Q15** If three numbers are in AP whose sum is 33 and product is 792, then the smallest number from these numbers is
 (A) 4 (B) 8
 (C) 11 (D) 14
- Q16** The sum of the first 100 terms common to the series $17, 21, 25, 29, 33, \dots$ and $16, 21, 26, 31, 36, \dots$ is
 (A) 100101 (B) 111000
 (C) 101100 (D) 110010
- Q17** The sum of integers from 1 to 100 that are divisible by 2 or 5 is equal to
 (A) 3000 (B) 3050
 (C) 4050 (D) None of these
- Q18** The number of terms common to the two sequences $17, 21, 25, \dots, 417$ and $16, 21, 26, \dots, 466$ is $4k$. Then, k is equal to
- Q19** If the sum of the 10 terms of an AP is 4 times to the sum of its 5 terms, then the ratio of first term and common difference is
 (A) 1 : 2 (B) 2 : 1
 (C) 2 : 3 (D) 3 : 2
- Q20** If the sides of a right angled triangle are in AP, then the sides are proportional to
 (A) 1 : 2 : 3
 (B) 2 : 3 : 4
 (C) 3 : 4 : 5
 (D) 4 : 5 : 6
- Q21** If a, b, c are distinct and are in G.P with common ratio r and $a, 2b, 3c$ are in AP then $r =$
 (A) 1
 (B) $\frac{1}{3}$
 (C) $-\frac{1}{3}$
 (D) -1



Answer Key

Q1 (C)
Q2 (A)
Q3 (B)
Q4 (C)
Q5 (B)
Q6 (B)
Q7 (B)
Q8 (D)
Q9 (C)
Q10 (D)
Q11 (C)

Q12 (B)
Q13 (C)
Q14 (D)
Q15 (A)
Q16 (C)
Q17 (B)
Q18 5
Q19 (A)
Q20 (C)
Q21 (B)



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